Principia mini spec

Centre-of-mass projection and invariant monitoring

Purpose. This note defines a lightweight runtime numerical policy for the planar three-body simulator. The atlas and decode pipeline already quotient out the centre-of-mass (COM) translation and bulk momentum modes at the initial-condition level. The runtime integrator should preserve that same reduced description by projecting the evolving state back onto the COM constraint manifold, while treating energy and angular momentum as monitored invariants rather than explicitly enforced constraints.

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1 Scope and design intent

The atlas already decodes trajectories into a COM-consistent representation. Runtime COM projection is therefore not a new gauge choice; it is numerical housekeeping that keeps floating-point drift from reintroducing the redundant modes that the atlas removed.

The design goal is to get the best precision-to-speed ratio available in an fp32 simulator:

- enforce COM position and total linear momentum exactly, or near-exactly, by projection,
- do *not* project energy or angular momentum,
- instead monitor the drift of physical invariants as diagnostics of numerical fidelity.

2 State and notation

For three bodies with masses m_i, positions $\mathbf{r}_i \in \mathbb{R}^2$, and either velocities $\mathbf{v}_i$ or momenta $\mathbf{p}_i = m_i\mathbf{v}_i$, define

$$M = m_1 + m_2 + m_3, \tag{1}$$

$$\mathbf{R}_{\text{com}} = \frac{1}{M}\sum_{i=1}^{3} m_i\mathbf{r}_i, \tag{2}$$

$$\mathbf{V}_{\text{com}} = \frac{1}{M}\sum_{i=1}^{3} m_i\mathbf{v}_i, \tag{3}$$

$$\mathbf{P}_{\text{com}} = \sum_{i=1}^{3} \mathbf{p}_i. \tag{4}$$

The intended reduced manifold is

$$\sum_{i=1}^{3} m_i\mathbf{r}_i = 0, \tag{5}$$

$$\sum_{i=1}^{3} \mathbf{p}_i = 0, \tag{6}$$

that is,

$$\mathbf{R}_{\text{com}} = 0, \tag{7}$$

$$\mathbf{P}_{\text{com}} = 0. \tag{8}$$

3 Why COM projection is special

COM position and total linear momentum are unusually cheap and safe to enforce because they are:

- linear constraints,
- generated by exact translation symmetry,
- dynamically decoupled from the internal relative motion,
- removable by a canonical projection with no ambiguity.

This is not true in the same way for energy or angular momentum. Restoring energy or L_z after drift requires choosing *how* to modify the state, and there is no unique, physically innocent correction in ordinary Cartesian coordinates. Such a correction would steer the orbit rather than merely remove a redundant gauge-like mode.

4 Runtime projection rule

4.1 Position and velocity form

If the integrator evolves positions and velocities, compute after each timestep (or at another chosen cadence)

$$\mathbf{R}_{\text{com}} = \frac{1}{M}\sum_{i=1}^{3} m_i \mathbf{r}_i, \tag{9}$$

$$\mathbf{V}_{\text{com}} = \frac{1}{M}\sum_{i=1}^{3} m_i \mathbf{v}_i, \tag{10}$$

and project back via

$$\mathbf{r}_i \leftarrow \mathbf{r}_i - \mathbf{R}_{\text{com}}, \tag{11}$$

$$\mathbf{v}_i \leftarrow \mathbf{v}_i - \mathbf{V}_{\text{com}}. \tag{12}$$

4.2 Position and momentum form

If the integrator evolves momenta instead, compute

$$\mathbf{R}_{\text{com}} = \frac{1}{M}\sum_{i=1}^{3} m_i \mathbf{r}_i, \tag{13}$$

$$\mathbf{P}_{\text{com}} = \sum_{i=1}^{3} \mathbf{p}_i, \tag{14}$$

and project via

$$\mathbf{r}_i \leftarrow \mathbf{r}_i - \mathbf{R}_{\text{com}}, \tag{15}$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i - \frac{m_i}{M}\mathbf{P}_{\text{com}}. \tag{16}$$

4.3 Recommended cadence

For a three-body fp32 browser simulator, projection every timestep is recommended by default. The cost is negligible compared with the integration itself, and the correction prevents precision from being wasted on spurious bulk drift.

5 Physical interpretation

Without runtime projection, the numerical state can drift into the form

$$\mathbf{r}_i = \mathbf{r}_i^{\text{internal}} + \mathbf{R}_{\text{drift}}, \tag{17}$$

where $\mathbf{R}_{\text{drift}}$ is physically irrelevant but grows in fp32 due to roundoff. As the absolute magnitude of the coordinates grows, the spacing between representable fp32 numbers grows too, so internal structure is resolved more poorly. COM projection removes this wasted dynamic range and keeps precision focused on the relative motion that actually matters.

6 What should not be enforced

The simulator should **not** explicitly project back to a target energy or target angular momentum during ordinary runtime evolution.

6.1 Energy

The total energy for Newtonian gravity is

$$E = \sum_{i=1}^{3} \frac{\|\mathbf{p}_i\|^2}{2m_i} - G \sum_{1 \le i < j \le 3} \frac{m_i m_j}{\|\mathbf{r}_i - \mathbf{r}_j\|}. \tag{18}$$

Enforcing $E = E_0$ after drift gives only one scalar condition. There are many possible state corrections satisfying that condition, and none is canonical in general. Explicit energy correction is therefore excluded from this runtime policy.

6.2 Angular momentum

In the planar problem,

$$L_z = \sum_{i=1}^{3} (x_i p_{y,i} - y_i p_{x,i}) = \sum_{i=1}^{3} m_i (x_i v_{y,i} - y_i v_{x,i}). \tag{19}$$

Again, forcing $L_z = L_{z,0}$ requires a non-unique modification of the state and can distort the true orbit. Explicit angular-momentum projection is therefore excluded from this runtime policy.

7 Invariant monitoring policy

Energy and angular momentum should instead be tracked as diagnostics. At minimum, record

$$\Delta E(t) = E(t) - E(0), \tag{20}$$

$$\Delta L_z(t) = L_z(t) - L_z(0). \tag{21}$$

A numerically robust relative form is also useful:

$$\delta_E(t) = \frac{E(t) - E(0)}{\max(\varepsilon_E, |E(0)|)}, \tag{22}$$

$$\delta_L(t) = \frac{L_z(t) - L_z(0)}{\max(\varepsilon_L, |L_z(0)|)}, \tag{23}$$

where ε_E and ε_L are small stabilising floors used when the initial invariant is close to zero.

7.1 Recommended diagnostics

The simulator should maintain and optionally expose the following quantities:

Quantity	Meaning	Runtime action
$\|\mathbf{R}_{\text{com}}\|$	COM position drift	project away
$\|\mathbf{P}_{\text{com}}\|$	COM momentum drift	project away
$\Delta E(t)$	absolute energy drift	monitor / log
$\Delta L_z(t)$	absolute angular-momentum drift	monitor / log
$\max_{\tau \le t} \lvert\Delta E(\tau)\rvert$	worst energy deviation so far	monitor / log
$\max_{\tau \le t} \lvert\Delta L_z(\tau)\rvert$	worst angular-momentum deviation so far	monitor / log

The *shape* of the drift matters:

- bounded oscillation is usually less concerning,
- steady secular drift is more concerning,
- sudden spikes often indicate close-encounter or timestep trouble.

8 Recommended implementation loop

A minimal runtime loop consistent with this policy is:

1. Decode an initial condition from the atlas.
2. Initialise the state in the COM frame.
3. Advance one timestep with the chosen integrator.
4. Recompute $\mathbf{R}_{\text{com}}$ and either $\mathbf{V}_{\text{com}}$ or $\mathbf{P}_{\text{com}}$.
5. Project the state back onto $\mathbf{R}_{\text{com}} = 0$ and $\mathbf{P}_{\text{com}} = 0$.
6. Evaluate diagnostic quantities $E(t)$ and $L_z(t)$.
7. Update drift histories and optional debug/UI outputs.

9 Practical notes

Notes

- COM projection is cheap enough to be treated as integrator housekeeping rather than a user-facing feature.
- This policy improves numerical efficiency, but it does not solve close-encounter stiffness, energy drift, or extreme escaper-vs-binary dynamic-range issues by itself.
- The method is especially well matched to Principia because the atlas already encodes the same reduced geometric viewpoint at the initial-condition level.

10 Spec summary

Runtime numerical policy.

- The simulator **shall** keep the evolving state in the centre-of-mass gauge by explicit runtime projection.
- The simulator **shall** remove COM position and bulk momentum drift after each timestep by a linear projection.
- The simulator **shall not** explicitly enforce energy or angular momentum by projection during ordinary evolution.
- The simulator **shall** track and expose energy and angular-momentum drift as diagnostics of numerical fidelity.